

Solving equations

- Solve an equation with fractions, ex. $\frac{7x}{8} - 2 = 3 - \frac{15x}{16}$.
 - Multiply by the lowest common denominator and solve.
- Solve an equation with absolute values, ex. $4|x + 1| - 7 = 6$.
 - Isolate the absolute value part, e.g., $|x + 1|$.
 - Solve for $\pm$ cases; Check your solutions.
- Solve an equation with multiple unknown values for a given variable, ex. $G = Q + (M - 7)P$ for M .
- Solve a n^{th} root, ex. $\sqrt{x-1} + 2 = 5$, $\sqrt[5]{x+1} = 4$.
 - Isolate the radical and raise both sides to the power of n .
 - Solve and Check your answers.

Solving quadratics

- Solve using the square root property, ex. $(x - 4)^2 + 1 = 2$.
 - Just solve like you do for x and treat the other letters like numbers.
 - Isolate the squared part, e.g., $(x - 4)^2$ and apply a square root to both sides; don't forget the $\pm$.
 - Solve and check your answers.
- Solve by factoring. ex. $3x^2 = -3x + 6$.
 - Set the equation equal to zero and factor.
 - Find the value for x that makes each factor 0.
- Solve using a graph.
 - Simply find the x coordinates when the graph intersects (or touches) the x -axis.
 - Note that this is an easy way to check the solutions for the above problems using your calculator.

The circle: $(x - h)^2 + (y - k)^2 = r^2$ where (h, k) is the center and r is the radius.

- Find the center and radius of a circle from the standard equation, ex. $(x - 2)^2 + (y + 4)^2 = 16$.
 - pick out your h, k , and r ; be careful with the $\pm$ signs.
- Write the standard equation of a circle for a given center and radius, ex. $(9, 4)$ and passing through the origin.
 - Fill in the equation. If you're given a point it passes through instead of the radius, you need to use the distance formula to find r .

Linear equations.

- Find the slope of a line going through two points (x_1, y_1) and (x_2, y_2) : $m = \frac{y_2 - y_1}{x_2 - x_1}$.
- Find the slope of a line perpendicular to $\ell_1 = m_1x + b_1$: $m = -\frac{1}{m_1}$, or parallel to $\ell_2 = m_2x + b_2$: $m = m_2$.
- Find the equation for a line: $y - y_1 = m(x - x_1) + b$ where the line passes through (x_1, y_1) .
- Graph a line.

Complex numbers.

- Perform $\times, \div, +, -$ with complex numbers (aka imaginary numbers), ex., $(2 - i)(4 + 8i)$.
- Write a complex number in standard form, i.e., $a + bi$, ex., write $\frac{4}{5 + i}$ as $\frac{5}{6} - \frac{1}{6}i$.
- Simply i^n , ex., $i^3 = -i$, $i^{40} = 1$, $i^1 = i$, $i^2 = -1$, $i^3 = -i$, and $i^4 = 1$. Write i^n in these terms, ex. $i^{42} = i^{40} \cdot i^2 = (i^4)^{10}(-1) = 1^{10}(-1) = -1$.

Functions.

- Given a relation is a function, ex., $g = \{(2, 7), (30, -4), (45, 7), (2, 5)\}$. Determine if it is a function, find its domain, and its range.
 - To determine if it is a function: Check that each x , in (x, y) , is paired with only one y value. If any x is paired with two different y 's then the relation is NOT a function.
 - The domain is all the x values, and the range is the y values.
- Given a graph, determine if it is a function, find its domain, and its range.
 - Use the vertical line test, ex. see Exam2 solutions #1.
- Use transformations to graph a function.
 - Know the parent functions of: $x^2, \sqrt{x}, a^x, \log_a x$.
 - Know the transformations $f(x + a) \leftarrow$; $f(x - a) \rightarrow$; $f(x) + a \uparrow$; $f(x) - a \downarrow$; $f(-x)$ y -axis reflection; $-f(x)$ x -axis reflection.
- Difference quotient of a $f(x)$, ex., find the difference quotient of $f(x) = 7x + 3$.
 - Use the formula, $\frac{f(x + h) - f(x)}{h}$.

- Find the average rate of change of a $f(x)$ from a to b , ex., what is the average rate of change of $f(x) = x^2 + 1$ from 0 to 2?
- Find the inverse of a given function, ex., $f(x) = -\frac{1}{3}x + 1$ find f^{-1} .
- Compute function computation, ex., let $f(x) = x + 4$ and $g(x) = x^2 - x$ find $(f + g)(x)$ and $(f \circ g)(2)$.

Quadratic functions.

- Find the vertex, axis of symmetry, max/min, range, domain, directrix and focus given a quadratic, ex., see #1 and #2 Exam2 and #9 Exam3.
 - First find the vertex, and then use it to answer the other parts.
 - If $a(x - h)^2 + k$: the vertex is (h, k) . If $ax^2 + bx + c$: the vertex is $(-b/2a, f(-b/2a))$.
- Find a quadratic given the equation OR the directrix and focus. ex., see #2 Exam3 and #10 Exam4.
 - Given the focus and directrix, $\frac{1}{4p}(x - h)^2 + k$ where p is $1/2$ the distance between the directrix and the focus. If the focus is (x, y) then the vertex is $(x, y - p)$.

Polynomials.

- Use traditional or synthetic division, ex., let $P(x) = x^2 - 5x + 2$
 - find the quotient and remainder, ex., find the quotient $Q(x)$ and remainder r when $P(x)$ is divided by $x - 2$.
 - Determine if a is a root of $P(x)$; a is a root if and only if $P(a) = 0$ or $P(x) \div (x - a)$ has no remainder.
 - Write $\frac{P(x)}{x - a}$, as a sum of polynomial and rational function: use the formula $\frac{P(x)}{x - a} = Q(x) + \frac{r}{x - a}$.
- Find the simplest polynomial with the given roots, ex., for roots $-2, 2, 5$ it is $(x + 2)(x - 2)(x - 5)$.
- Find the x -ints of a polynomial; set the function equal to 0 and solve.
- Determine if the x -ints pass or touches the x -axis.
 - look at the multiplicity: if it is even it touches, if it is odd it passes.
- Find all vertical and horizontal asymptotes.
 - Vertical: where the function is undefined (usually when it is 0 in the denominator), ex., $\frac{2x + 6}{x - 2}$ has vertical asymptote at $x = 2$ since this makes the denominator 0.
 - Horizontal: exists if the limit of the function converges, ex., $\lim_{x \rightarrow \infty} \frac{2x + 6}{x - 2} \cdot \frac{1/x}{1/x} = \frac{2 + 6/x}{1 - 2/x} = 2$ so $\frac{2x + 6}{x - 2}$ has horizontal asymptote 2.

Limits.

- Determine y as $x \rightarrow \pm\infty$, ex., $x^2 - 20$ as $x \rightarrow -\infty$
- Determine $\lim_{x \rightarrow a^\pm} f(x)$, for polynomial, rational, exponential, and logarithmic functions, ex., $\lim_{x \rightarrow \infty} \frac{1}{x^3}$ and $\lim_{x \rightarrow 2^-} \frac{3}{x - 2}$
 - $\lim_{x \rightarrow \infty} x = \infty$ and $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$, $\lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty$, and $\lim_{x \rightarrow 0^+} \frac{1}{x} = \infty$

Exponential and Logarithmic functions.

- Evaluate exponents, ex., -2^{-3} , $16^{2/4}$, $\ln(e^5)$, $\log_b 1$.
 - Need to be familiar with the rules of exponents and logs.
- Find the graph, domain, and range given a log or exponential function.
- Use properties of logs to condense an expression, ex., $\ln 2 + \ln x^6 = \ln(2x^6)$.
- Use the properties of logs to expand an expression, ex., $\log\left(\frac{10^6}{x}\right) = 6 - \log x$.
- Solve an exponential function, ex., $1500 = 500e^{0.10 \cdot t}$ for t .
 - Use a $\log_b x$, usually $\ln x$, on BOTH sides ex., $\ln 1500 = \ln 500e^{0.10 \cdot t}$
- Solve a logarithmic function, ex., $\ln(40) = 3 - \ln(x)$ for x .
 - Use an exponent of the same base as the logs on BOTH sides, ex., $\log_6 3x = 2 \Rightarrow 6^{\log_6 3x} = 6^2 \Rightarrow 3x = 36$.